

Dataset — Collection, Composition & Annotation

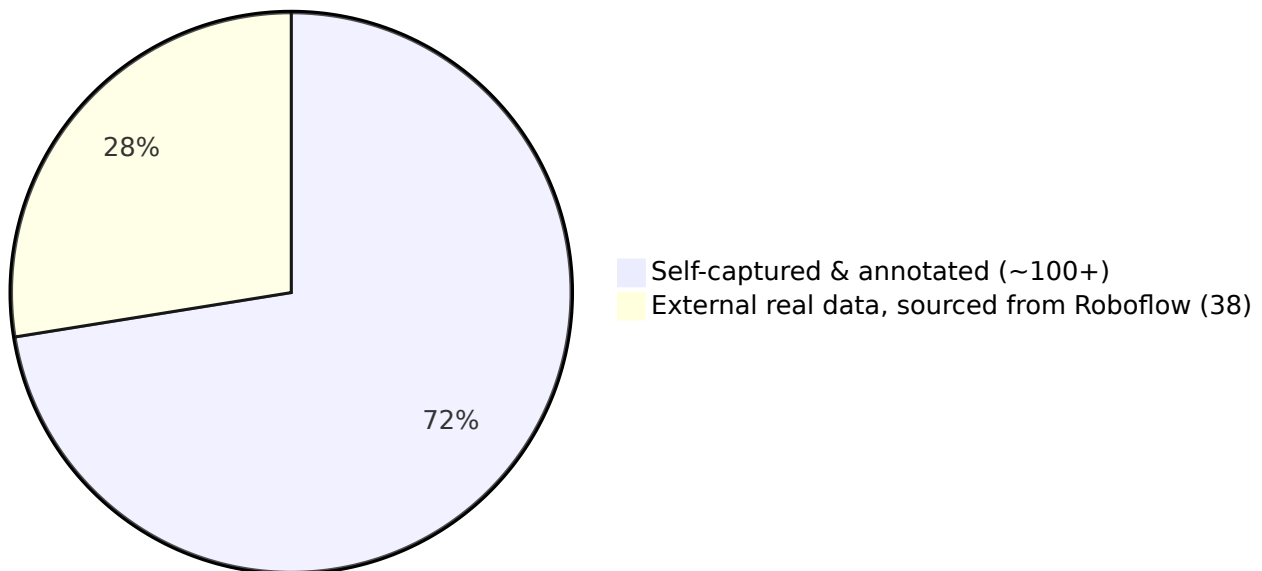
1. Data at a glance

Metric	Value
Total images captured	500
Locations	Hostel reading rooms + Library
Capture device	Mobile phone camera
Images manually reviewed & annotated	100+ (subset of the 500, due to 48-hour time constraint)
Desk instances labeled	400+
Annotation method	Fully manual, image-by-image inspection
Labeling scheme	Each desk labeled; if an object/person is present on the desk, labeled accordingly
Augmentation	Applied to the annotated subset before training
Models trained	2× YOLOv8 (fine-tuned separately)
Real-world test	Live mobile video (proxy for CCTV feed)
External real data used	38 images, sourced from a public Roboflow dataset (real, not synthetic)
External data share of training data	38 / (100+ self-captured + 38 external) $\approx$ 27% of the training set
Synthetic (fake/augmented-only) data used	None — no synthetic data in the final training set

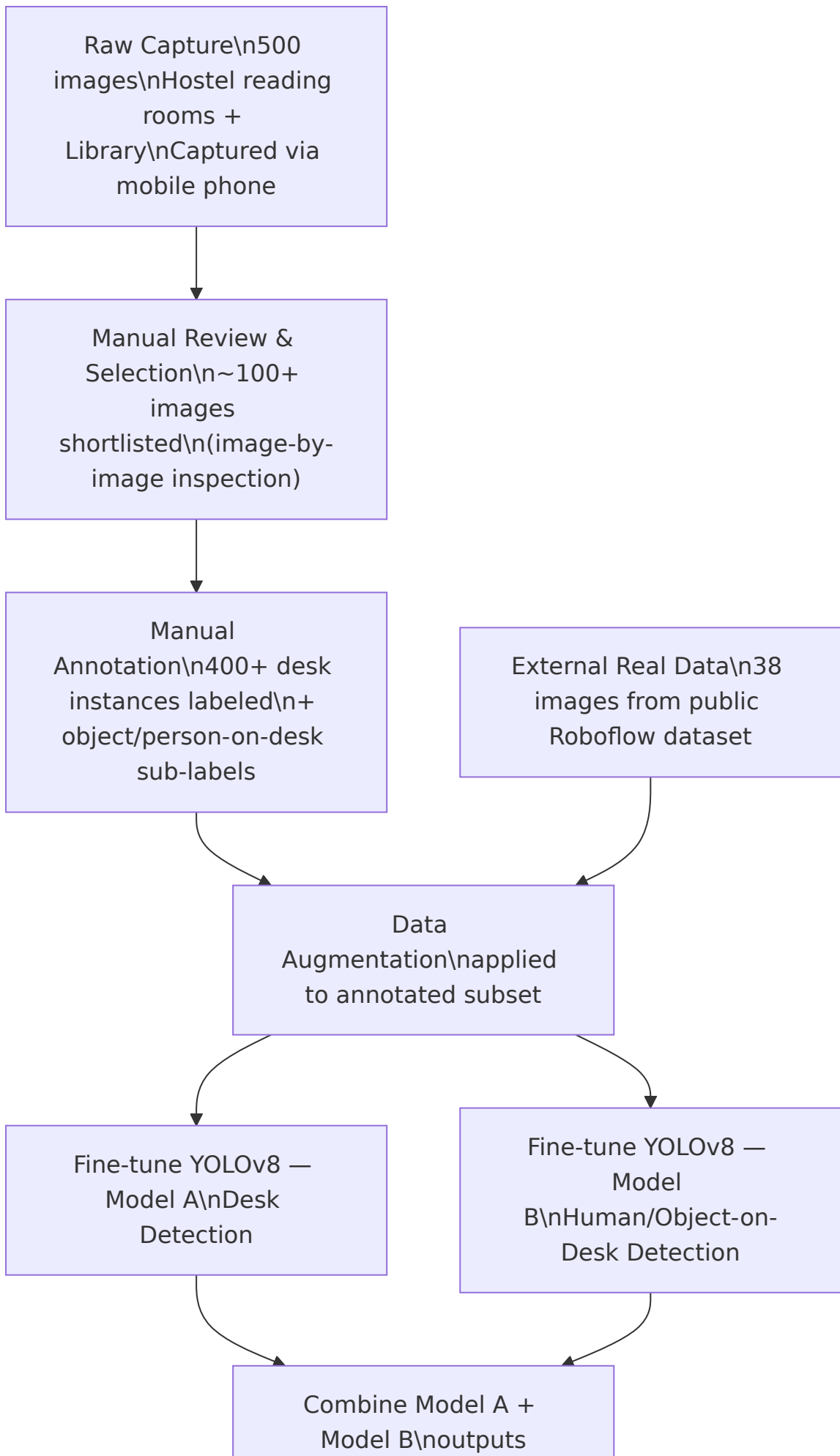
Honesty note: Of the 500 images captured, only 100+ (~20%) were manually annotated and used for training due to the hackathon's time constraint. The remaining ~400 images were captured but not processed — kept as raw evidence of the collection effort, not used for model training. To supplement this, 38 real images from a public Roboflow dataset were added to the training set (~27% of total training images) — self-captured data remains the majority and the foundation of the dataset; the external data extends it, it does not replace it. No synthetic (artificially generated) data was used in the final training set.

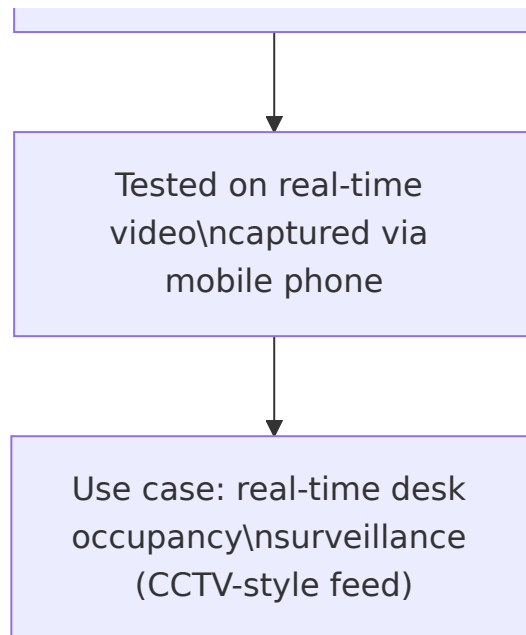
2. Training set composition

| Set Composition: Self-captured vs External



3. Data flow — capture to real-time inference





4. Observed vs. Inferred vs. Synthetic

Type	Description
Observed (self-captured)	The 500 raw images captured directly from hostel reading rooms and the library via mobile phone; the real-time test video captured the same way.
Observed (external, real)	38 real images sourced from a public Roboflow dataset, added to the training set (~27% of total training images) — not self-captured, but not synthetic either.
Inferred	Desk occupancy state (vacant / occupied / object-present) as predicted by the combined YOLOv8 models at inference time.
Synthetic	None used in the final training set.

5. How AI was used

- **YOLOv8** fine-tuned as two separate models:
 1. Desk detection (localizing desks in a frame)
 2. Human/object-on-desk detection (identifying what's occupying a desk)
- Outputs from both models combined to produce a per-desk occupancy status.
- Tested on live mobile video to simulate a real-time CCTV surveillance feed.

6. Known gaps & limitations (honesty > polish)

- Only ~20% of captured images were annotated and used for training, due to the 48-hour constraint — the rest remain raw/unprocessed.
- Manual annotation is subject to individual labeling consistency; no second-pass inter-annotator check was done.
- The 38 Roboflow-sourced images come from a different environment than our hostel/library setting — possible domain gap (lighting, desk style, camera angle) versus our own captures.